

Geometry

Unit 11

Lesson 4 Homework

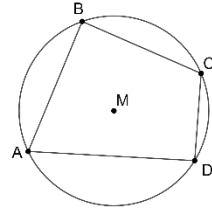
Name **Answer Key**

Date _____

Period _____

Circle M is shown, with $ABCD$ inscribed in the circle.

The $m\widehat{ADC} = 184^\circ$, and $m\angle DAB = 44^\circ$.



1. Find the $m\angle ABC$.

$$\frac{184}{2} = 92^\circ; m\angle ABC = 92^\circ$$

2. Find the $m\angle BCD$.

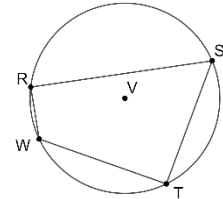
$$m\angle BCD = 180 - 44 = 136^\circ$$

3. Find the $m\angle CDA$.

$$m\angle CDA = 180 - 92 = 88^\circ$$

Circle V is shown, with $RSTW$ inscribed in the circle.

The $m\widehat{WTS} = 190^\circ$, and $m\angle WTS = (5x + 10)^\circ$.



4. Find the value of x .

$$\frac{190}{2} = 95^\circ; 180 - 95 = 85 = m\angle WTS$$

$$5x + 10 = 85$$

$$5x = 75$$

$$x = 15$$

5. Find the $m\angle WTS$.

$$m\angle WTS = 85^\circ$$

Circle B is shown, with $GHJK$ inscribed in the circle. The $m\angle HGK = (5x + 25)^\circ$, and $m\angle GKJ = (3x)^\circ$ and $m\angle KJH = (4x + 20)^\circ$.

6. Find the value of x .

$$5x + 25 + 4x + 20 = 180$$

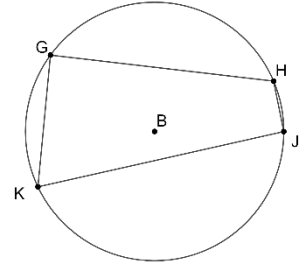
$$9x + 45 = 180$$

$$9x = 135$$

$$x = 15$$

7. Find $m\angle HGK$.

$$m\angle HGK = 5(15) + 25 = 100^\circ$$



8. Find $m\angle GKJ$.

$$m\angle GKJ = 3(15) = 45^\circ$$

9. Find $m\angle KJH$.

$$m\angle KJH = 4(15) + 20 = 80^\circ$$

10. Find $m\angle JHG$.

$$m\angle JHG = 360 - (100 + 45 + 80) = 135^\circ$$